

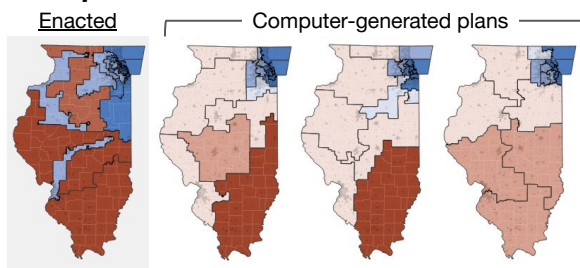
How algorithms can benchmark redistricting

Computer-drawn redistricting plans can be a **nonpartisan baseline** which can be used to assess the plans adopted by commissions and legislatures. These algorithms can be **tailored** to use different nonpartisan criteria and then **transparently** show if and where gerrymandering is occurring.

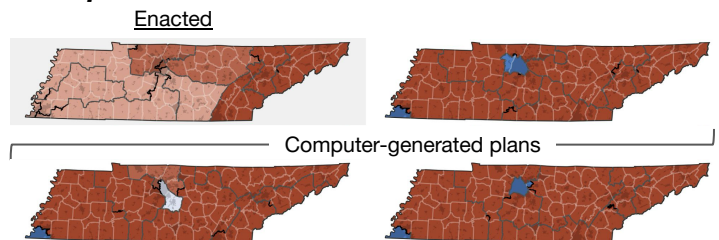
1 Comparing different states or time periods can't detect or measure gerrymandering.
Every state has its own demographics, partisanship, geography, and redistricting rules, which change over time. This makes it hard to predict how many seats each party should win, even with fair districts. For example, Maryland and Colorado have plans drawn by commissions and similar statewide results, but Maryland's delegation is **7D–1R** while Colorado's delegation is **4D–4R**.

2 Computer algorithms can draw thousands of redistricting plans that satisfy traditional criteria like population balance and compactness.
Algorithms are transparent because they use only specified criteria to draw redistricting plans. They can also be used to assess how different criteria matter and their tradeoffs. The examples below prioritize geographically compact districts that avoid splitting counties and municipalities.

Example: Illinois

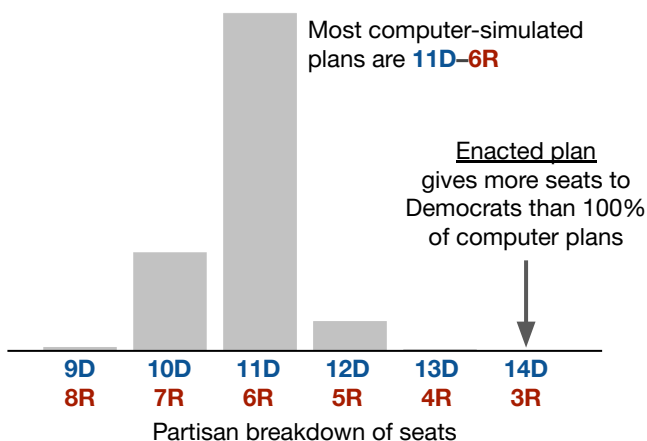


Example: Tennessee



3 Comparing the actual districts to computer-drawn ones can indicate gerrymandering.
Big gaps between enacted and computer-drawn districts can indicate partisan or other manipulations.

Comparison: Illinois



Comparison: Tennessee

